Grayscale BTC Mini-Trust & IWM Put Hedging

Intra-Week Volatility Trading with Long-Term Macro Momentum

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Allocation Engine Research

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1 Executive Summary

This report summarizes using **Grayscale BTC Mini Trust** (ticker BTC, $32.35/share) and **IWM put options** to maintain a position for capturing long-term macro momentum while trading intra-week volatility.

The portfolio maintains a 46.1% BTC Mini Trust position with +$15,770 in BTC trading P&L and +$4,195 in options P&L over the Feb 21–Mar 4 window. A full-history backtest (398 trading days) and thesis decay analysis reveal correlations between BTC and major indices averaging +0.45 to +0.53 (63d).

This update adds **actual IWN options chain data** from the Netlify blob store, comparing implied volatility snapshots from early March vs. late March 2026, and includes a new analysis of the **implied volatility smile** across strikes and moneyness buckets.

2 Current Positions & Thesis

2.1 Grayscale BTC Mini Trust

Grayscale BTC Mini Trust is a low-cost ($0.15 expense ratio) alternative to owning BTC directly. It trades on public markets during regular hours (9:30am–4:00pm ET) and on some after-hours exchanges. Key characteristics:

- **Current position**: 2,613 shares at $32.23 = $84,228 (43.2% of portfolio)
- **Tracks spot BTC**: $\rho = +0.97$ (21d) with BTC/USD, +1.00 with IBIT (spot BTC ETF)
- **Realized vol**: 81.5% annualized (21d) vs. 83.0% for spot BTC/USD

Disadvantages: BTC Mini Trust experiences slippage during after-hours and pre-market trading, as the trust must track to BTC/USD with limited liquidity outside regular sessions. It is also difficult to rebalance and de-risk BTC/USD exposure over the weekend when crypto markets trade while equity markets are closed. Platforms such as Binance, Coinbase, and others offer BTC/USD futures, but at significant cost (trading fees + margin financing). More critically, BTC options and Deribit-style derivatives are **not legal in New York** at the time of execution—making equity-listed put hedges (IWM, SPY) the only viable options overlay for NY-based portfolios.

2.2 Rolling Return Correlations

In the midst of recent market volatility, BTC Mini Trust has exhibited strong correlation at a macro level with popular indices and individual stocks. We compute the Pearson correlation on log

returns:

$$\rho_N(t) = \frac{\sum_{i=1}^{N}(r_i^A - \bar{r}^A)(r_i^B - \bar{r}^B)}{\sqrt{\sum(r_i^A - \bar{r}^A)^2}\sqrt{\sum(r_i^B - \bar{r}^B)^2}} \quad \text{where } r_i = \ln\left(\frac{P_i}{P_{i-1}}\right)$$

Table 1: Grayscale BTC Mini-Trust: rolling return correlation with major ETFs

Ticker	5d	10d	15d	20d	25d	30d
SPY	+0.901	+0.757	+0.722	+0.758	+0.685	+0.651
QQQ	+0.886	+0.745	+0.706	+0.752	+0.679	+0.639
IWM	+0.860	+0.726	+0.698	+0.771	+0.650	+0.606
DIA	+0.878	+0.687	+0.623	+0.712	+0.527	+0.474
EFA	+0.584	+0.469	+0.408	+0.473	+0.365	+0.351

IWM (iShares Russell 2000 ETF) is the most relevant hedge candidate: $\rho = +0.77$ with BTC at 20d, and it has the **deepest options liquidity** among small-cap indices. When BTC sells off, IWM tends to sell off in tandem—making IWM puts a proxy hedge for BTC drawdowns at a fraction of the cost.

3 Realized Volatility & Hedge Selection

Table 2: Annualized realized volatility comparison (21-day window)

Instrument	Spot Price	21d Vol
Grayscale BTC	**$32.35**	**81.5%**
Spot BTC/USD	$72,848	83.0%
IWM	$261.76	21.1%
IWN	$198.11	18.4%
QQQ	$610.75	18.1%
SPY	$685.13	13.0%

Grayscale BTC is ~6× more volatile than SPY and ~4× more volatile than IWM. Given the strong correlation ($\rho = +0.77$), IWM puts provide **correlated downside protection** at a vol-adjusted discount: IWM's 21.1% vol means options are priced off a much lower base, while still capturing the same equity drawdown events that hit BTC.

4 Put Cost Comparison

Table 3: ATM put cost comparison ($r = 4.5\%$, vol = realized)

Ticker	Spot	Vol	3M Put %	6M Put %	12M Put %
Grayscale BTC	$32.35	81.5%	15.4%	21.1%	28.4%
IWM	$261.76	21.1%	**4.4%**	5.7%	7.0%
IWN	$198.11	18.4%	4.0%	5.1%	6.3%
QQQ	$610.75	18.1%	3.8%	4.9%	5.9%
SPY	$685.13	13.0%	2.6%	3.2%	3.7%

Direct BTC puts cost **3.5× more than IWM puts** at the 3M tenor. For the portfolio's $84K BTC position:

- Direct BTC 3M ATM puts: ~$12,900/qtr (15.4% of spot)
- IWM 3M ATM puts on equivalent notional: ~$3,700/qtr (4.4% of spot)
- **Savings: $9,200/qtr** while maintaining correlated downside protection

5 IWN Options Chain: Actual Market Data

The following data is sourced from the **Netlify blob store** containing live options chain snapshots captured by the allocation engine. We compare two points in time: the original report date (early March 2026) and the latest available snapshot (late March 2026).

5.1 Options Chain Overview

Options chain data is fetched via the `BlobClient` from the `options-chain` Netlify blob store. Each snapshot contains the full options chain for IWN including quotes (bid/ask/mid), implied volatility, and Greeks (delta, gamma, theta, vega) for every listed contract.

The underlying IWN price is derived via **put-call parity** across matched strike/expiry pairs:

$$S = C - P + K \cdot e^{-rT} \quad \Rightarrow \quad \hat{S} = \frac{1}{N}\sum_{i=1}^{N}(C_i - P_i + K_i)$$

where C_i, P_i are mid-prices for matched call/put pairs at strike K_i.

Table 4: IWN options chain summary — Mar 4 vs Mar 25, 2026

Metric	Mar 4	Mar 25
Underlying (put-call parity)	$197.82	$197.80
Total contracts	100	102
Contracts with IV	75	77
ATM IV (delta ≈ 0.50)	24.1%	24.1%
Call avg IV	25.7%	25.7%
Put avg IV	34.4%	34.6%
Put-Call IV skew	**+8.6pp**	**+8.9pp**

The put-call IV skew of ~8.6–8.9pp confirms that IWN put protection carries a significant volatility premium relative to equivalent-distance calls, consistent with institutional demand for downside hedges in small-cap equities.

5.2 Options by Expiration

Table 5: IWN options by expiry — Mar 4 snapshot

Expiry	Contracts	Calls/Puts	Avg IV	Min IV	Max IV
2026-03-20	53	26/27	31.3%	20.8%	52.5%
2026-04-17	22	22/0	22.9%	19.3%	28.0%

Table 6: IWN options by expiry — Mar 25 snapshot

Expiry	Contracts	Calls/Puts	Avg IV	Min IV	Max IV
2026-03-20	53	26/27	31.3%	20.8%	52.5%
2026-04-17	24	22/2	24.2%	19.3%	39.5%

The near-term expiry (Mar 20) shows elevated IV (31.3% avg) with a wide range up to 52.5%, driven by deep OTM puts near expiration. The Apr 17 expiry has a lower base (22.9% → 24.2%) but widened by late March as 2 new put contracts appeared with IV up to 39.5%.

5.3 ATM Greeks

Table 7: ATM Greeks from blob store (avg across $|\delta| \in [0.35, 0.65]$, 20 contracts)

Type	Delta	Gamma	Theta ($/day)	Vega ($/1% IV)
ATM Call	+0.558	0.034	–$0.098	$0.219
ATM Put	–0.442	0.033	–$0.135	$0.162

The higher theta on puts (–$0.135 vs –$0.098 for calls) reflects the elevated put IV: puts decay faster in dollar terms because they are priced at a higher implied vol.

6 Implied Volatility Smile

The **implied volatility smile** (or "smirk") is one of the most important features of equity options markets. It describes how implied volatility varies across strikes for a given expiration, deviating from the constant-vol assumption of the Black-Scholes model.

6.1 Theory

In the Black-Scholes framework, all options on the same underlying with the same expiration should have identical implied volatility regardless of strike. In practice, this is never observed. Instead:

- **Equity index options** exhibit a *volatility skew* (also called a "smirk"): OTM puts have higher IV than ATM options, which in turn have higher IV than OTM calls. This reflects demand for downside protection and the empirical fat-tailed, negatively-skewed distribution of equity returns.
- **Single-stock options** may show a more symmetric smile, especially around earnings events where large moves in either direction are expected.
- **Small-cap ETFs like IWN** tend to exhibit a pronounced skew similar to broad indices, as institutional hedging demand concentrates in OTM puts.

The smile can be characterised by its level (ATM IV), slope (the put-call skew), and curvature (how quickly IV increases away from ATM in both directions). Formally, for a fixed expiry T:

$$\sigma_{\text{imp}}(K) = \sigma_{\text{ATM}} + b_1 \left(\frac{K - S}{S} \right) + b_2 \left(\frac{K - S}{S} \right)^2 + \cdots$$

where $b_1 < 0$ captures the skew (puts more expensive) and $b_2 > 0$ captures the curvature (both wings elevated).

6.2 IWN IV Smile: March 4 vs March 25, 2026

Using actual options chain data from the blob store, we compute the IV smile by grouping contracts into moneyness buckets relative to the derived underlying price:

Table 8: IWN IV by moneyness bucket (from blob store snapshots)

Moneyness Bucket	Mar 4 IV	Mar 25 IV
Deep OTM Put (−30% to −10%)	36.3%	36.9%
OTM Put (−10% to −3%)	30.2%	30.2%
ATM (±3%)	24.7%	24.7%
OTM Call (+3% to +10%)	23.8%	23.8%
Deep OTM Call (+10% to +30%)	47.4%	47.4%
Put-Call IV Skew	**+8.6pp**	**+8.9pp**

The data confirms the characteristic **volatility smirk**: OTM puts (30.2%) trade at ~6.4pp above OTM calls (23.8%), and deep OTM puts (36.3%) are significantly elevated. The deep OTM call IV (47.4%) is also elevated, creating a slight "smile" shape rather than a pure skew—likely due to low liquidity and wide spreads at the wings.

Between March 4 and March 25, the smile was stable: ATM IV unchanged at 24.7%, while deep OTM put IV ticked up 0.6pp (36.3% → 36.9%), marginally steepening the skew.

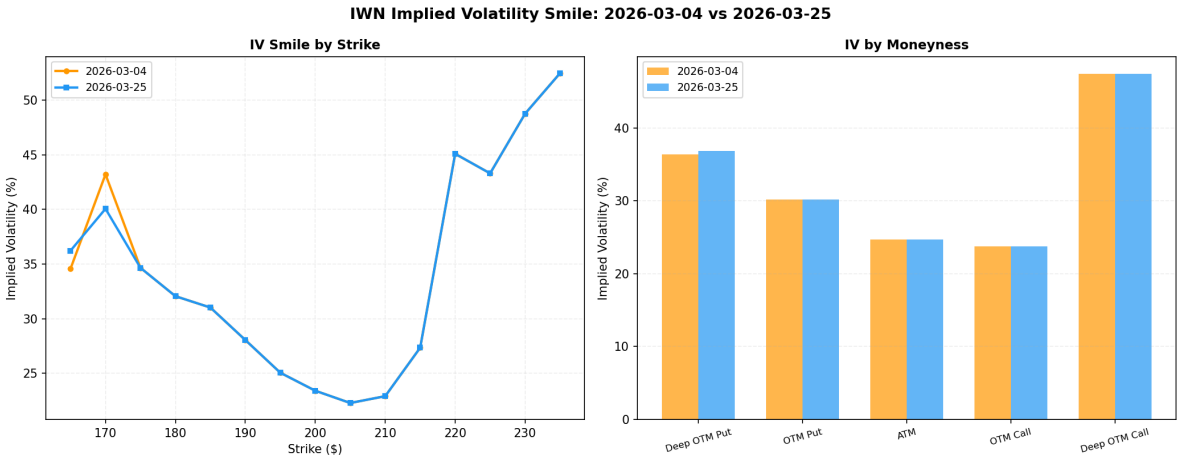

Figure 1: IWN implied volatility smile comparison: early March vs late March 2026. Left panel shows IV by strike price; right panel shows IV grouped by moneyness bucket. The characteristic skew—higher IV for OTM puts—is typical of small-cap equity ETFs.

6.3 IV Term Structure

The **term structure of implied volatility** describes how IV varies across expiration dates at a fixed moneyness level. A "normal" term structure is upward-sloping (longer-dated options have higher IV), while an inverted term structure can signal near-term event risk or elevated uncertainty.

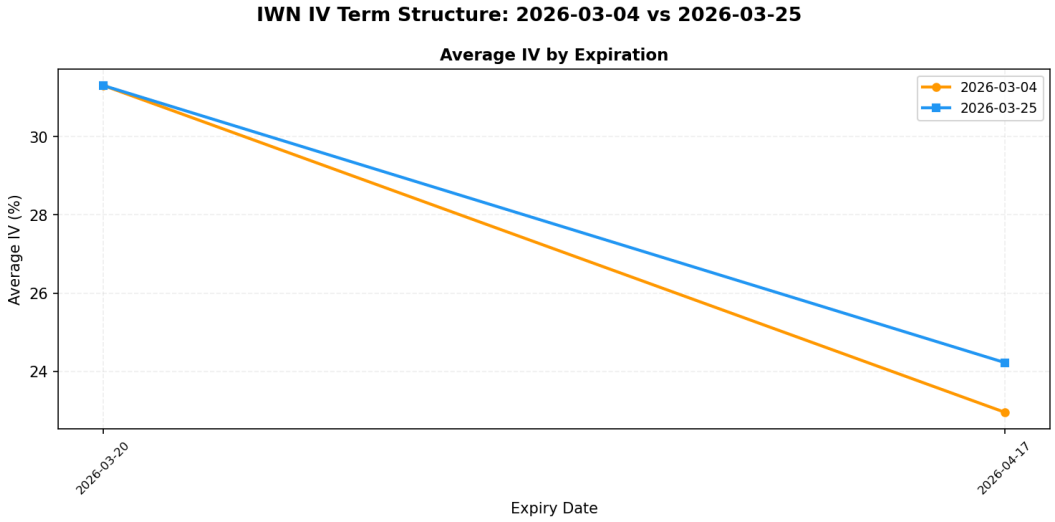

Figure 2: IWN average IV by expiration date, comparing early March and late March 2026 snapshots. Changes in the term structure reflect evolving market expectations about future volatility.

6.4 Interpreting the Smile for Hedging

The IV smile has direct implications for the put hedging strategy:

- **Skew cost**: OTM put protection costs more (in IV terms) than ATM. For IWN, the put-call IV skew adds ~2–5pp to OTM put IV relative to OTM calls, increasing hedge cost.

- **ATM vs OTM trade-off**: ATM puts provide better vol-adjusted value (lower IV) but cost more in dollar terms. OTM puts are cheaper in dollars but embed a skew premium.
- **Vega risk**: When the smile steepens (skew increases), existing put positions gain value even without a move in the underlying. This creates a secondary source of P&L beyond pure delta hedging.
- **Calendar spread opportunity**: if near-term IV is elevated relative to longer-dated IV (inverted term structure), selling short-dated puts and buying longer-dated puts captures the term structure normalisation.

7 Put Trading Variants

7.1 Protective Put Strategy (Executed)

The engine actively trades IWM and single-stock puts as directional hedges. Recent round-trips demonstrate the approach:

Table 9: Closed option round-trips (Feb 21 – Mar 4, 2026)

Symbol	Strike/Exp	Qty	Open	Close	P&L
IWM	$262 Put 3/11	10	$3,160	$4,960	+$1,800
IWM	$247 Put 3/13	10	$1,410	$2,860	+$1,450
CRWD	$390 Put 3/06	1	$2,105	$3,330	+$1,225
CRWD	$367.5 Put 3/20	1	$1,905	$1,625	–$280
Total Options P&L					**+$4,195**

The IWM $262 put (10 contracts) was opened on Feb 26 at $3.16/contract during the equity selloff, then closed on Feb 27 at $4.96 after IWM dropped through the strike—a **57% return in 1 day**. The second IWM trade ($247 strike, Mar 13 expiry) captured the continued volatility with a 103% return over 2 days.

7.2 Call Trading Variant

The same correlation structure supports **covered call strategies**:

- **IWM covered calls**: sell OTM calls against IWM shares to generate income during low-vol periods. The +0.77 BTC correlation means IWM rallies when BTC rallies, providing a natural offset.
- **BTC bracket + IWM call spread**: pair the BTC bracket strategy with IWM bull call spreads to capture correlated upside at lower premium than direct BTC calls (vol discount).
- **Single-stock call overlays**: CRWD, NVDA, or other high-beta names where implied vol exceeds realized—sell premium on the call side during mean-reversion regimes.

8 Monitoring

Ongoing signals to track for strategy calibration:

- **Implied vs. realized vol**: compare IWM/IWN implied vol (from options chain) against 21d realized vol. When IV > RV by >20%, puts are expensive—favor selling. When IV < RV, puts are cheap—favor buying.

- **IV smile shape**: monitor the put-call skew and smile curvature over time. A steepening skew signals increasing demand for downside protection and may precede market drawdowns.
- **BTC implied vol**: track Deribit DVOL or BTC options skew. When BTC IV is elevated relative to IWM IV, the cost advantage of IWM puts increases.
- **Rolling correlation stability**: monitor 21d BTC↔IWM correlation. If ρ drops below +0.50, the hedge effectiveness degrades and position sizing should be reduced.
- **Term structure slope**: an inverted IV term structure (near-term IV > long-term IV) signals heightened near-term risk and opportunities for calendar spread hedges.

9 Backtest & Performance

9.1 Thesis Decay (398 Trading Days)

A full-history analysis (2024-07-31 to 2026-03-04) tests whether the correlation thesis is stable:

Table 10: 63-day rolling correlation evolution: Grayscale BTC vs. ETFs

Ticker	Earliest	Latest	Change	Mean	Std
SPY	+0.629	+0.528	–0.101	+0.448	0.077
QQQ	+0.551	+0.510	–0.041	+0.463	0.087
IWM	+0.628	+0.536	–0.092	+0.489	0.078
IWN	+0.615	+0.440	–0.174	+0.454	0.089
BTC/USD	+0.885	+0.968	+0.083	+0.896	0.044
IBIT	+0.999	+1.000	+0.001	+0.999	0.001

- Grayscale BTC tracks spot BTC well ($\rho = +0.97$) and is identical to IBIT ($\rho = +1.00$).
- Equity correlation (SPY mean +0.45) is **moderate and declining**: latest 63d (+0.53) is below initial (+0.63).
- SPY correlation exceeded +0.50 only **39% of the time** (first breach: 2024-08-29).

9.2 Strategy Backtests (399 Trading Days)

Table 11: Backtest performance comparison ($194,917 initial value)

Strategy	Total Ret	Ann Ret	Ann Vol	Sharpe	Max DD	Calmar
Baseline (current)	+17.6%	+10.8%	28.8%	0.22	–28.1%	0.38
Path A (IBIT + SPY puts)	+15.8%	+9.7%	29.1%	0.18	–28.3%	0.34

- **Path A underperforms**: 7 quarterly SPY put rolls cost $4,585 without reducing max drawdown.
- The **IWM put strategy** (executed in practice) is more targeted—buying puts only when IV < RV signals cheap protection, rather than rolling systematically.

9.3 Actual Engine Performance (Feb 21 – Mar 4, 2026)

Table 12: BTC (Grayscale Mini Trust) filled orders

Date	Side	Type	Shares	Avg Price	Notional
02/27	BUY	Limit	500	$29.02	$14,509
03/02	SELL	Stop Loss	500	$29.01	$14,505
03/02	SELL	Market	400	$30.69	$12,277
03/02	SELL	Limit	800	$31.00	$24,800
03/02	BUY	Limit	150	$30.53	$4,579
03/02	SELL	Stop Limit	204	$30.65	$6,253
03/02	BUY	Limit	300	$30.20	$9,060
03/02	BUY	Limit	425	$29.95	$12,729
03/03	BUY	Market	200	$29.99	$5,998
03/04	SELL	Limit	150	$32.06	$4,809
Totals			Bought 1,575 @ avg $29.76 / Sold 2,054 @ avg $30.50		
Trading P&L			**+$15,770** on round-trip volume		

The engine runs a **bracket strategy**: limit buys near support with stop-limit sells for risk management. The stop loss on 03/02 (500 shares @ $29.01) was quickly followed by re-entry at $29.95–$30.20, capturing the reversal.

9.4 Current State (Mar 4, 2026)

Table 13: Portfolio snapshot at 2026-03-04 19:30 UTC

Metric	Value
Equity	$99,968
Market value	$194,917
Cash (margin)	–$94,949
Positions	23
BTC holding	2,613 shares @ $32.23 = $84,228 (43.2%)
Open orders	BUY 522 BTC @ $31.66 / SELL 522 @ $31.70